

NATIONAL UNIVERSITY OF SINGAPORE

MA3236 - Nonlinear Programming

Midterm Test

(Semester 1 : AY2023/2024)

Date: 04 Oct 2023, Time: 12:15pm - 1:30pm

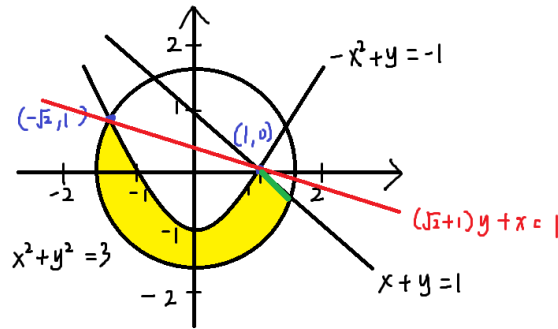
INSTRUCTIONS TO CANDIDATES

1. Please write your matriculation/student number only. Do not write your name.
2. This examination paper contains **5** questions and comprises **2** printed pages.
3. Answer **ALL** questions.
4. The total marks for this paper is **50**.
5. Please start each question on a new page.
6. This is a CLOSED BOOK examination.
7. Students are allowed to use one double-sided, A4 size helpsheet.
8. Candidates may use scientific calculators. Graphical and programmable calculators are not allowed. However, they should lay out systematically the various steps in the calculations.

1. [10 marks] Consider the following problem (P):

$$\begin{aligned}
 \text{(P)} \quad & \min_{x,y \in \mathbb{R}} \quad \alpha x - y \\
 \text{s.t.} \quad & -x^2 + y \leq -1 \\
 & x + y \leq 1 \\
 & x^2 + y^2 \leq 3
 \end{aligned}$$

- (a) Draw out the feasible region of (P).
 (b) Using the graph in (a), determine α so that (P) has
 (i) infinitely many solutions
 (ii) exactly two solutions
 (c) Determine the optimal objective function value for each case in (b).



Solution.

From the graph, (P) has infinitely many solutions on the green part. In this case, the objective function is parallel to $x + y = 1$ or $-x - y = -1$. To make the function decreases upwards, we set $\alpha = -1$, and hence the optimal objective function value is -1.

On the other hand, (P) has exactly two solutions when both $(-\sqrt{2}, 1)$ and $(1, 0)$ are its solution. The red line passing through the two points is

$$-\frac{1}{\sqrt{2}+1}x - y = -\frac{1}{\sqrt{2}+1}.$$

To make the function decreases upwards, we set $\alpha = -\frac{1}{\sqrt{2}+1}$, and hence the optimal objective function value is $-\frac{1}{\sqrt{2}+1}$.

2. [6 marks] Let C_1 and C_2 be convex sets in $\mathbb{R}^{m+p}$, and let C be the set of vectors $\mathbf{x} = (\mathbf{y}; \mathbf{z})$ (where $\mathbf{y} \in \mathbb{R}^m$ and $\mathbf{z} \in \mathbb{R}^p$) such that there exist vectors $\mathbf{z}_1$ and $\mathbf{z}_2$ with $(\mathbf{y}; \mathbf{z}_1) \in C_1$, $(\mathbf{y}; \mathbf{z}_2) \in C_2$ and $\mathbf{z}_1 + \mathbf{z}_2 = \mathbf{z}$. Prove that C is a convex set in $\mathbb{R}^{m+p}$.

Solution.

Take $\mathbf{x} = (\mathbf{y}; \mathbf{z})$, $\mathbf{x}' = (\mathbf{y}'; \mathbf{z}')$ $\in C$.

We have $(\mathbf{y}; \mathbf{z}_1), (\mathbf{y}'; \mathbf{z}'_1) \in C_1$, $(\mathbf{y}; \mathbf{z}_2), (\mathbf{y}'; \mathbf{z}'_2) \in C_2$ for some $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}'_1, \mathbf{z}'_2$ such that $\mathbf{z}_1 + \mathbf{z}_2 = \mathbf{z}$, $\mathbf{z}'_1 + \mathbf{z}'_2 = \mathbf{z}'$.

For any $\lambda \in [0, 1]$,

$$\begin{aligned} \lambda \mathbf{x} + (1 - \lambda) \mathbf{x}' &= \begin{pmatrix} \lambda \mathbf{y} + (1 - \lambda) \mathbf{y}' \\ \lambda \mathbf{z} + (1 - \lambda) \mathbf{z}' \end{pmatrix} \\ &= \begin{pmatrix} \lambda \mathbf{y} + (1 - \lambda) \mathbf{y}' \\ \lambda(\mathbf{z}_1 + \mathbf{z}_2) + (1 - \lambda)(\mathbf{z}'_1 + \mathbf{z}'_2) \end{pmatrix} \\ &= \begin{pmatrix} \lambda \mathbf{y} + (1 - \lambda) \mathbf{y}' \\ \lambda \mathbf{z}_1 + (1 - \lambda) \mathbf{z}'_1 + \lambda \mathbf{z}_2 + (1 - \lambda) \mathbf{z}'_2 \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned} \begin{pmatrix} \lambda \mathbf{y} + (1 - \lambda) \mathbf{y}' \\ \lambda \mathbf{z}_1 + (1 - \lambda) \mathbf{z}'_1 \end{pmatrix} &= \lambda(\mathbf{y}; \mathbf{z}_1) + (1 - \lambda)(\mathbf{y}'; \mathbf{z}'_1) \in C_1, \\ \begin{pmatrix} \lambda \mathbf{y} + (1 - \lambda) \mathbf{y}' \\ \lambda \mathbf{z}_2 + (1 - \lambda) \mathbf{z}'_2 \end{pmatrix} &= \lambda(\mathbf{y}; \mathbf{z}_2) + (1 - \lambda)(\mathbf{y}'; \mathbf{z}'_2) \in C_2, \end{aligned}$$

since C_1 and C_2 are convex. By definition, this also shows that C is convex.

3. [10 marks] Suppose we have a coercive function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and a convex function $h : \mathbb{R}^n \rightarrow \mathbb{R}$. Define $(g + h)(\mathbf{x}) := g(\mathbf{x}) + h(\mathbf{x})$.

- (a) Can we conclude that $g + h$ is always coercive?
- (b) Can we conclude that $g + h$ is always convex?

Justify your answer for each question.

Solution.

- (a) No. Take $g(x) = |x|$ and $h(x) = 2x$. Then

$$(g + h)(x) = \begin{cases} 3x, & \text{if } x \geq 0, \\ x, & \text{otherwise.} \end{cases}$$

So $g + h \rightarrow -\infty$ as $x \rightarrow -\infty$.

- (b) No. Take $g(x) = x^2(x^2 - 1) = x^4 - x^2$ and $h(x) = x$. Then $(g + h)(x) = x^4 - x^2 + x$, $(g + h)'(x) = 4x^3 - 2x + 1$, $(g + h)''(x) = 12x^2 - 2$ is not nonnegative for every $x \in \mathbb{R}$. In particular, $(g + h)''(0) = -2 < 0$.

Note: sum of two nonconvex functions can be convex, eg. $x^3 - x^3 = 0$.

4. [14 marks] Consider $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}) = x_1^3 - 5x_1^2 - 8x_1 + x_2^2 + x_3^2 - 2x_2x_3$$

- (a) Find all the stationary points of f in $\mathbb{R}^3$.
- (b) For each stationary point obtained in (a), determine whether it is a local maximizer or a local minimizer or a saddle point of f in $\mathbb{R}^3$.
- (c) Does f have a global minimizer in $\mathbb{R}^3$? Justify your answer.

Solution.

(a)

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 3x_1^2 - 10x_1 - 8 \\ 2x_2 - 2x_3 \\ 2x_3 - 2x_2 \end{bmatrix} = \mathbf{0}$$

From the second and third equation, $x_2 = x_3$. From the first equation,

$$(3x_1 + 2)(x_1 - 4) = 0 \Leftrightarrow x_1 = -\frac{2}{3} \text{ or } x_1 = 4$$

So the stationary points in $\mathbb{R}^3$ are

$$\left\{ \left[-\frac{2}{3}; t; t \right] : t \in \mathbb{R} \right\} \cup \left\{ [4; t; t] : t \in \mathbb{R} \right\}$$

(b)

$$H_f(\mathbf{x}) = \begin{bmatrix} 6x_1 - 10 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{bmatrix}.$$

The eigenvalue of this matrix is $6x_1 - 10$, 4, 0, where 4, 0 are the eigenvalue of the submatrix $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$.

When $x_1 = -2/3$, $6x_1 - 10 = -14$, so the Hessian matrix is indefinite. The set of stationary points $\left\{ \left[-\frac{2}{3}; t; t \right] : t \in \mathbb{R} \right\}$ are all saddle points.

When $x_1 = 4$, $6x_1 - 10 = 14$, so the Hessian matrix is positive semidefinite. We cannot use necessary condition to make conclusion. However, note that

$$f(\mathbf{x}) = x_1^3 - 5x_1^2 - 8x_1 + (x_2 - x_3)^2 \geq x_1^3 - 5x_1^2 - 8x_1 := g(x_1).$$

Since $x_1 = 4$ is a local minimizer of g , and (t, t) is a global minimizer of $(x_2 - x_3)^2$ for each $t \in \mathbb{R}$, $(4, t, t)$ is a local minimizer of f for all $t \in \mathbb{R}$.

(c) No. As $x_1 \rightarrow -\infty$, $f \rightarrow -\infty$.

5. [10 marks] Consider $\min \left\{ f(x) := \frac{1}{x^2} + \frac{1}{4}(x-1)^2 \mid x > 0 \right\}$.

(a) Show that f has exactly one global minimizer on the set $D := \{x \in \mathbb{R} \mid x > 0\}$.

- (b) Manually run two iterations of Newton's method to obtain x_2 , starting with the initial guess $x_0 = 1$ to solve $\min_{x>0} f(x)$.

Solution.

- (a) $f'(x) = -\frac{2}{x^3} + \frac{1}{2}x - \frac{1}{2}$, $f''(x) = \frac{6}{x^4} + \frac{1}{2} > 0$ implies f is strictly convex. If x^* is a stationary point on D , x^* is the unique global minimizer of f on D .

Note that $f'(1) < 0$, $f'(2) > 0$. By the intermediate value theorem, f' has a root in $\mathbb{R}$, and that root is the unique global minimizer.

Remarks. Here, we need to show both existence and uniqueness of global minimizer. Note that f is strictly convex does not mean it has global minimizer, eg. $f(x) = e^{-x}$. Also, f is coercive on $\mathbb{R}$ cannot be used to show the existence of global minimizer in this question, since the domain is $\mathbb{R}_+$.

- (b) We have $f'(x) = -\frac{2}{x^3} + \frac{1}{2}x - \frac{1}{2}$, $f''(x) = \frac{6}{x^4} + \frac{1}{2}$. Then

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = 1 - \frac{-2}{13/2} = \frac{17}{13} = 1.3077.$$

$$x_2 = x_1 - \frac{f'(x_1)}{f''(x_1)} = 1.5979.$$

End of paper